

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Examination – Winter 2024

Course: B. Tech. Branch: Mechanical Engineering/Mechanical Engg(Sandwitch) Semester: VII

Subject Code & Name: BTHM702-Industrial Engineering & Management

Max Marks: 60

Date: 07/02/2025

Duration: 3 Hr.

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 is compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

**Q. 1 Objective type questions. (Compulsory Question)**

**12**

- |   |                                                                                                                                                                                         |     |   |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|
| 1 | Who of the following is not related to the contribution in scientific management?                                                                                                       | CO1 | 1 |
|   | a. F. W. Taylor      b. Henri Fayol      c. Elton Mayo      d. Louis Pasteur                                                                                                            |     |   |
| 2 | Planning is a process by which _____                                                                                                                                                    | CO1 | 1 |
|   | a. manager select, train, promote their staff      b. a manager anticipates the future      c. manager achieve harmony of individual efforts      d. manager monitor program activities |     |   |
| 3 | Staffing is a process by which _____                                                                                                                                                    | CO2 | 1 |
|   | a. a manager anticipates the future      b. manager select, train, promote their staff      c. manager achieve harmony of individual efforts      d. manager monitor program activities |     |   |
| 4 | _____ is the quality of the behavior of the managers whereby they inspire confidence and trust in their subordinates and guide their activities in organized efforts.                   | CO2 | 1 |
|   | a. leadership      b. planning      c. staffing      d. forecasting                                                                                                                     |     |   |
| 5 | Which of the following is not a sales forecasting technique?                                                                                                                            | CO3 | 1 |
|   | a. Market Survey      b. Delphi Method      c. Correlation Analysis      d. Batch Production                                                                                            |     |   |
| 6 | Which of the following term is not related to Material Requirements Planning (MRP)                                                                                                      | CO3 | 1 |
|   | a. Economic order quantity      b. Bill of Materials file      c. Inventory record file      d. Matching people to jobs                                                                 |     |   |
| 7 | The facilities layout in which similar machines or similar operations are kept at one location is called as _____                                                                       | CO4 | 1 |
|   | a. Product Layout      b. Combination Layout      c. Process Layout      d. None of these                                                                                               |     |   |
| 8 | The layout in which the men and equipment are moved to the                                                                                                                              | CO4 | 1 |

|                                                            |                                                                                                                                                            |                                   |                                           |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------|
|                                                            | material, which remains at one place is called as,                                                                                                         |                                   |                                           |
|                                                            | a. Product Layout                                                                                                                                          | b. Fixed position layout          | c. Process Layout                         |
|                                                            |                                                                                                                                                            |                                   | d. None of these                          |
| 9                                                          | By clearly stating the goals of the business and meshing them with the goals of the employees, managers make use of a management technique called as ..... |                                   | CO4 1                                     |
|                                                            | a. Total Quality Management (TQM)                                                                                                                          | b. Management by Objectives (MBO) | c. Total Preventive Maintenance (TPM)     |
|                                                            |                                                                                                                                                            |                                   | d. All of these                           |
| 10                                                         | _____ is concerned with investigating, reducing and eliminating ineffective time, whatever may be the cause                                                |                                   | CO5 1                                     |
|                                                            | a. Product Layout                                                                                                                                          | b. Work measurement               | c. Bill of Materials file                 |
|                                                            |                                                                                                                                                            |                                   | d. Economic order quantity                |
| 11                                                         | Motion studies are used to _____                                                                                                                           |                                   | CO5 1                                     |
|                                                            | a. Develop the best work method                                                                                                                            | b. Develop motion consciousness   | c. Develop economical and efficient tools |
|                                                            |                                                                                                                                                            |                                   | d. All of these                           |
| 12                                                         | Which of the following is not a Total Quality Management technique?                                                                                        |                                   | CO5 1                                     |
|                                                            | a. Just in time (JIT)                                                                                                                                      | b. quality circles                | c. six sigma                              |
|                                                            |                                                                                                                                                            |                                   | d. Bill of Materials                      |
| <b>Q. 2 Solve the following.</b>                           |                                                                                                                                                            |                                   |                                           |
| A)                                                         | What is management? Explain nature, characteristics and objectives of management.                                                                          | CO1                               | 12 6                                      |
| B)                                                         | Explain various advantages of decentralization of authority.                                                                                               | CO1                               | 6                                         |
| <b>Q. 3 Solve the following.</b>                           |                                                                                                                                                            |                                   |                                           |
| A)                                                         | Explain the importance of performance appraisal and career strategy.                                                                                       | CO2                               | 12 6                                      |
| B)                                                         | Briefly explain how team building, and communication help managers in effective leadership.                                                                | CO2                               | 6                                         |
| <b>Q. 4 Solve Any Two of the following.</b>                |                                                                                                                                                            |                                   |                                           |
| A)                                                         | Explain any one type of facility layout in detail.                                                                                                         | CO3                               | 12 6                                      |
| B)                                                         | What are the different characteristics of services systems?                                                                                                | CO3                               | 6                                         |
| C)                                                         | What is materials requirement planning?                                                                                                                    | CO3                               | 6                                         |
| <b>Q. 5 Write Short Notes on Any Two of the following.</b> |                                                                                                                                                            |                                   |                                           |
| A)                                                         | Capacity planning                                                                                                                                          | CO4                               | 12 6                                      |
| B)                                                         | Facilities layout                                                                                                                                          | CO4                               | 6                                         |
| C)                                                         | Assembly line balancing                                                                                                                                    | CO4                               | 6                                         |
| <b>Q. 6 Write Short Notes on Any Two of the following.</b> |                                                                                                                                                            |                                   |                                           |
| A)                                                         | Micro motion study                                                                                                                                         | CO5                               | 12 6                                      |
| B)                                                         | Basic principles of ergonomics                                                                                                                             | CO5                               | 6                                         |
| C)                                                         | Quality circles                                                                                                                                            | CO5                               | 6                                         |

\*\*\* End \*\*\*

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Winter 2024

Course: B. Tech. Branch: Mechanical Engineering/ Mechanical Engineering(Sandwich)

Subject Code & Name: BTMEC704B-Industrial Engineering & Management Semester: VII

Max Marks: 60

Date:12/02/2025

Duration: 3 Hr.

**Instructions to the Students:**

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(Level/CO) Marks

**Q. 1 Objective type questions. (Compulsory Question)**

**12**

- |   |                                                                                                                                                                                         |     |   |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|
| 1 | Who of the following is related to the contribution in scientific management?                                                                                                           | CO1 | 1 |
|   | a. Bill Gates      b. Henri Fayol      c. Graham Bell      d. Louis Pasteur                                                                                                             |     |   |
| 2 | Planning is a process by which _____                                                                                                                                                    | CO1 | 1 |
|   | a. manager select, train, promote their staff      b. a manager anticipates the future      c. manager achieve harmony of individual efforts      d. manager monitor program activities |     |   |
| 3 | Staffing is a process by which _____                                                                                                                                                    | CO2 | 1 |
|   | a. a manager anticipates the future      b. manager select, train, promote their staff      c. manager achieve harmony of individual efforts      d. manager monitor program activities |     |   |
| 4 | _____ is the quality of the behavior of the managers whereby they inspire confidence and trust in their subordinates and guide their activities in organized efforts.                   | CO2 | 1 |
|   | a. leadership      b. planning      c. staffing      d. forecasting                                                                                                                     |     |   |
| 5 | Which of the following is not a sales forecasting technique?                                                                                                                            | CO3 | 1 |
|   | a. Market Survey      b. Delphi Method      c. Correlation Analysis      d. Batch Production                                                                                            |     |   |
| 6 | Which of the following term is not related to Material Requirements Planning (MRP)                                                                                                      | CO3 | 1 |
|   | a. Economic order quantity      b. Bill of Materials file      c. Inventory record file      d. Matching people to jobs                                                                 |     |   |
| 7 | The facilities layout in which similar machines or similar operations are kept at one location is called as _____                                                                       | CO4 | 1 |
|   | a. Product Layout      b. Combination Layout      c. Process Layout      d. None of these                                                                                               |     |   |
| 8 | The layout in which the men and equipment are moved to the                                                                                                                              | CO4 | 1 |

|                                                            |                                                                                                                                                            |                                   |                                           |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------|
|                                                            | material, which remains at one place is called as,                                                                                                         |                                   |                                           |
|                                                            | a. Product Layout                                                                                                                                          | b. Fixed position layout          | c. Process Layout                         |
|                                                            |                                                                                                                                                            |                                   | d. None of these                          |
| 9                                                          | By clearly stating the goals of the business and meshing them with the goals of the employees, managers make use of a management technique called as ..... |                                   | CO4 1                                     |
|                                                            | a. Total Quality Management (TQM)                                                                                                                          | b. Management by Objectives (MBO) | c. Total Preventive Maintenance (TPM)     |
|                                                            |                                                                                                                                                            |                                   | d. All of these                           |
| 10                                                         | _____ is concerned with investigating, reducing and eliminating ineffective time, whatever may be the cause                                                |                                   | CO5 1                                     |
|                                                            | a. Product Layout                                                                                                                                          | b. Work measurement               | c. Bill of Materials file                 |
|                                                            |                                                                                                                                                            |                                   | d. Economic order quantity                |
| 11                                                         | Motion studies are used to _____                                                                                                                           |                                   | CO5 1                                     |
|                                                            | a. Develop the best work method                                                                                                                            | b. Develop motion consciousness   | c. Develop economical and efficient tools |
|                                                            |                                                                                                                                                            |                                   | d. All of these                           |
| 12                                                         | Which of the following is not a Total Quality Management technique?                                                                                        |                                   | CO5 1                                     |
|                                                            | a. Just in time (JIT)                                                                                                                                      | b. quality circles                | c. six sigma                              |
|                                                            |                                                                                                                                                            |                                   | d. Bill of Materials                      |
| <b>Q. 2 Solve the following.</b>                           |                                                                                                                                                            |                                   |                                           |
| A)                                                         | What are the different functions of Management?                                                                                                            | CO1                               | 12 6                                      |
| B)                                                         | Explain the concept of decentralization in the organization.                                                                                               | CO1                               | 6                                         |
| <b>Q. 3 Solve the following.</b>                           |                                                                                                                                                            |                                   |                                           |
| A)                                                         | Explain the importance of performance appraisal and career strategy.                                                                                       | CO2                               | 12 6                                      |
| B)                                                         | Briefly explain how team building, and communication help managers in effective leadership.                                                                | CO2                               | 6                                         |
| <b>Q. 4 Solve Any Two of the following.</b>                |                                                                                                                                                            |                                   |                                           |
| A)                                                         | Explain any one type of facility layout in detail.                                                                                                         | CO3                               | 12 6                                      |
| B)                                                         | What are the different characteristics of services systems?                                                                                                | CO3                               | 6                                         |
| C)                                                         | What is materials requirement planning?                                                                                                                    | CO3                               | 6                                         |
| <b>Q. 5 Write Short Notes on Any Two of the following.</b> |                                                                                                                                                            |                                   |                                           |
| A)                                                         | Capacity planning                                                                                                                                          | CO4                               | 12 6                                      |
| B)                                                         | Facilities layout                                                                                                                                          | CO4                               | 6                                         |
| C)                                                         | Assembly line balancing                                                                                                                                    | CO4                               | 6                                         |
| <b>Q. 6 Write Short Notes on Any Two of the following.</b> |                                                                                                                                                            |                                   |                                           |
| A)                                                         | Micro motion study                                                                                                                                         | CO5                               | 12 6                                      |
| B)                                                         | Concurrent engineering                                                                                                                                     | CO5                               | 6                                         |
| C)                                                         | Quality circles                                                                                                                                            | CO5                               | 6                                         |

\*\*\* End \*\*\*